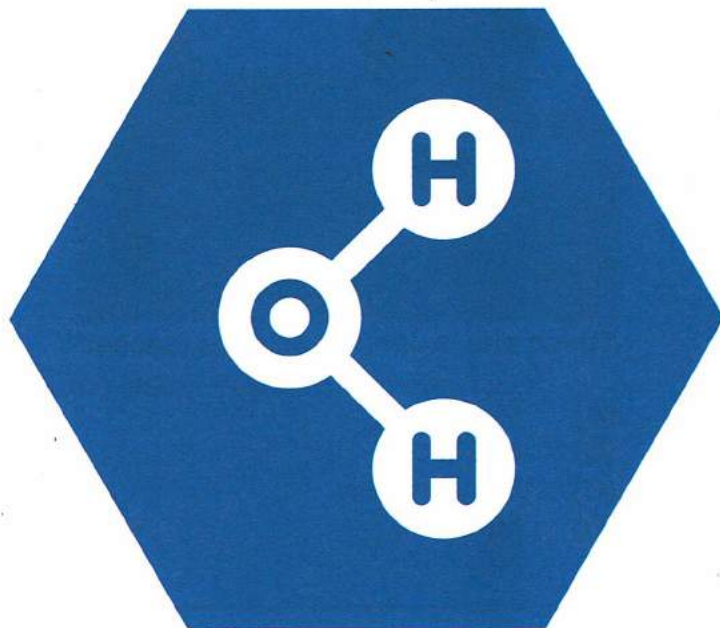




UNSW Global
AUSTRALIA

PAPER B



2016 ICAS

International Competitions
and Assessments for Schools

SCIENCE

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED.

30 QUESTIONS

TIME ALLOWED: 45 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.

Use a 2B or B pencil.

Do **NOT** use a pen.

Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.

Your score will be the number of correct answers.

Marks are **NOT** deducted for incorrect answers.

There are **30 MULTIPLE-CHOICE QUESTIONS** (1–30).

Use the information provided to choose the **BEST** answer from the four possible options.

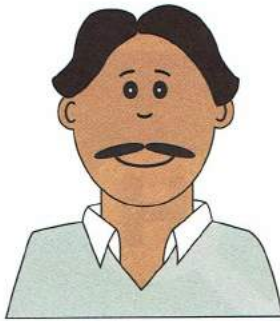
On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a calculator and a ruler.

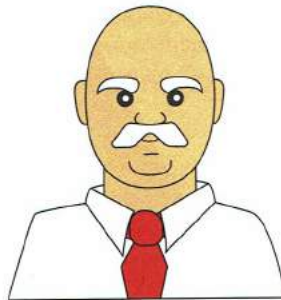
1. Jenny was describing her science teacher to her friends. She said her teacher has a beard, brown hair and he wears glasses.

The drawings show the four science teachers at Jenny's school.

Mr Auburn



Mr Bloom



Ms Conneely



Mr Duffin



Which one is Jenny's science teacher?

- (A) Mr Auburn
- (B) Mr Bloom
- (C) Ms Conneely
- (D) Mr Duffin

2. All mammals have two to four different types of teeth. For example, humans have four types of teeth for different jobs: canines, molars, premolars and incisors.

The diagram shows four skulls.

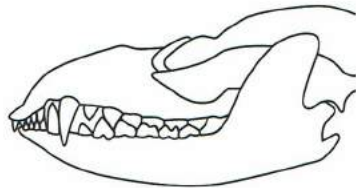


Figure 1

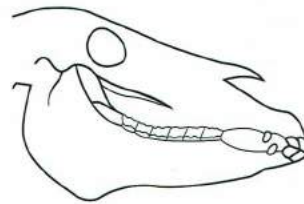


Figure 2

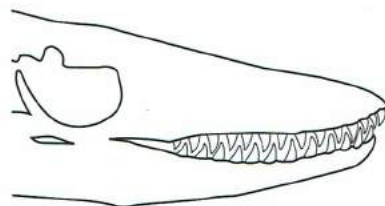


Figure 3

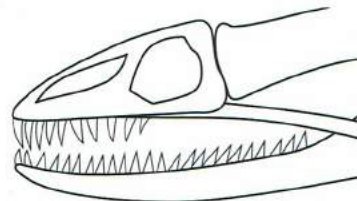


Figure 4

Which of the skulls belong to mammals?

- (A) Figure 1 and 2 only
- (B) Figure 2 only
- (C) Figure 1, 3 and 4 only
- (D) Figure 3 and 4 only

3. Some bodies of water on Earth's surface, such as oceans, rivers and lakes, are connected to each other. Water can be moved from one place to another by tides, evaporation, wind and rain.

Ruisi took two photos of the river outside his house, one in the morning and one in the afternoon.

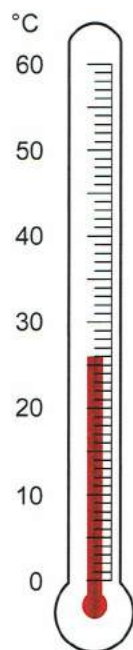


What is the most likely reason for the change of the water level in the photos?

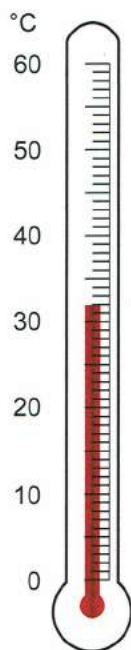
- (A) The river pours into a lake.
- (B) There is a drain next to the river.
- (C) Water evaporates on a very hot day.
- (D) The river is affected by tides in an ocean.

4. Adam measured the room temperature and the temperature of a glass of water.

room
temperature



water
temperature



What is the difference between the room temperature and the water temperature?

- (A) 5 °C
- (B) 6 °C
- (C) 7 °C
- (D) 10 °C

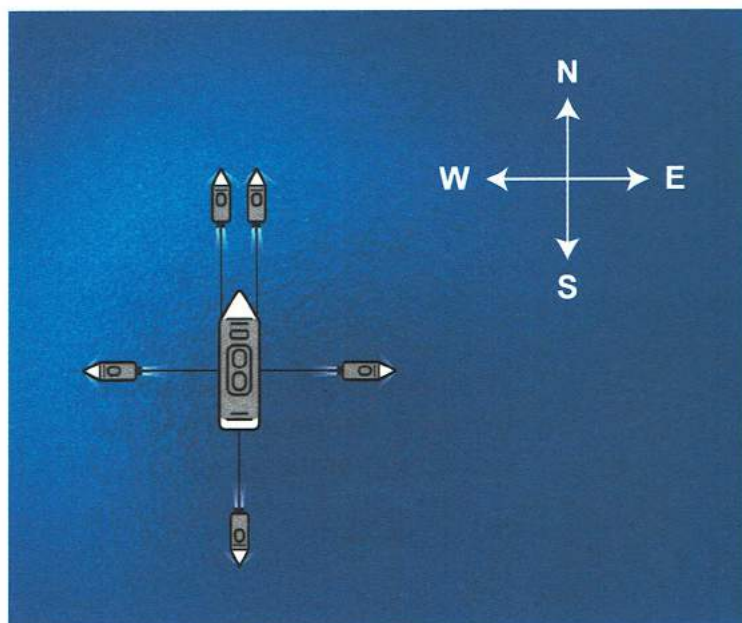
For questions 5 and 6 use the information below.

The table lists some information about tigers. They are endangered animals, with not many left in the wild or in zoos.

	Type of tiger			
	Siberian	Bengal	Indochinese	Sumatran
Body mass (kg)	90 – 300	136 – 210	115 – 180	90 – 120
Where it lives	Russia	India	Thailand	Sumatra
Number in the wild	360 – 400	3200 – 4600	1200 – 1800	400 – 500
Number in zoos	490	330	60	210

5. Where do the largest tigers live?
- (A) Russia (B) India (C) Thailand (D) China
6. Which two types of tiger have fewer than 2000 living in the wild but more than 100 living in zoos?
- (A) Indochinese and Sumatran (B) Siberian and Indochinese
(C) Siberian and Bengal (D) Siberian and Sumatran

7. The diagram shows a number of tugboats pulling a ship. All forces applied by the tugboats are equally strong.

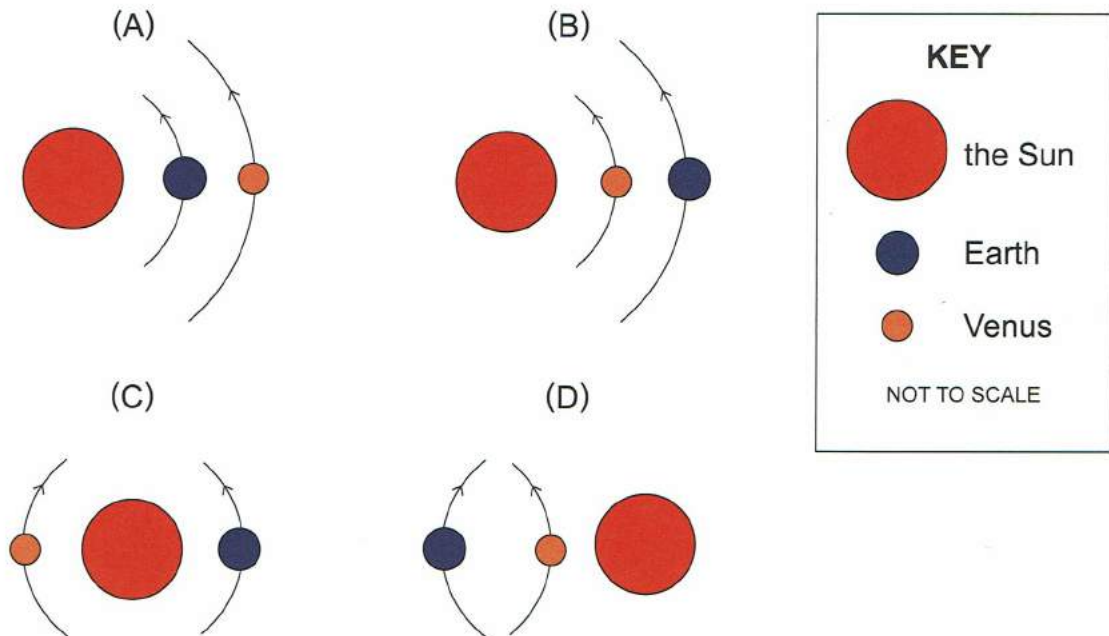


In which direction will the ship move?

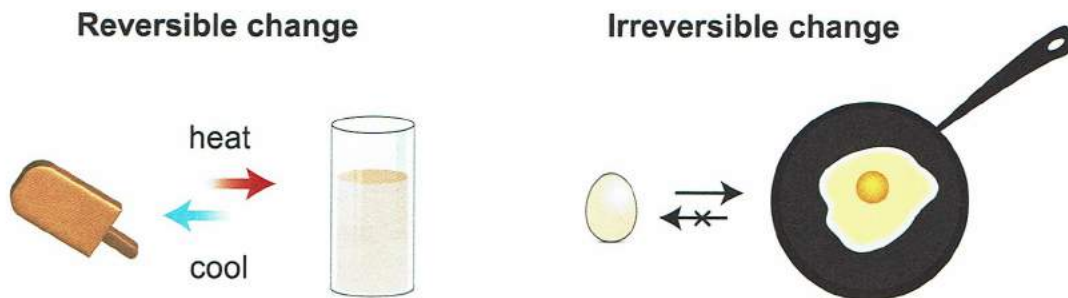
- (A) North
(B) East
(C) South
(D) West

8. The transit of Venus occurs when the planet Venus travels between the Sun and Earth. It can be observed from Earth.

Which diagram correctly shows the position of the Sun, Venus and Earth when a transit of Venus takes place?



9. Some changes can be easily undone. These changes are called reversible changes. Changes that cannot be undone are called irreversible changes.



Which of the following is a reversible change?

- (A) green leaves turning yellow in autumn
- (B) a piece of metal being pressed into coins
- (C) a piece of apple going brown
- (D) paper burning to ashes

10. Some students took these photographs.



spider



damselfly

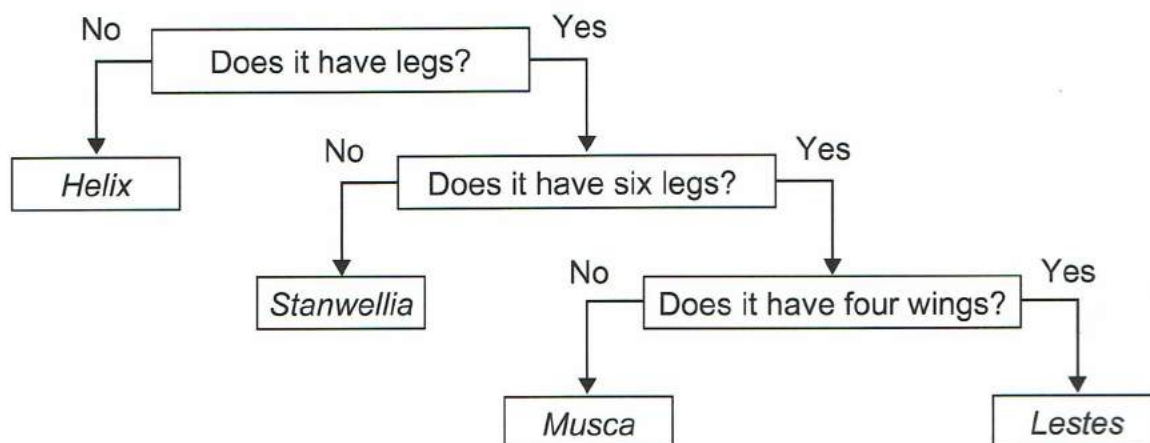


fly



snail

They used the key below to find the scientific name of the fly.



What is the scientific name of the fly?

- (A) *Helix* (B) *Stanwellia* (C) *Musca* (D) *Lestes*

11. When an object is dropped from a height, gravity causes it to fall to the ground. Sam took two identical sheets of paper. He crushed one sheet of paper and left the other one flat.

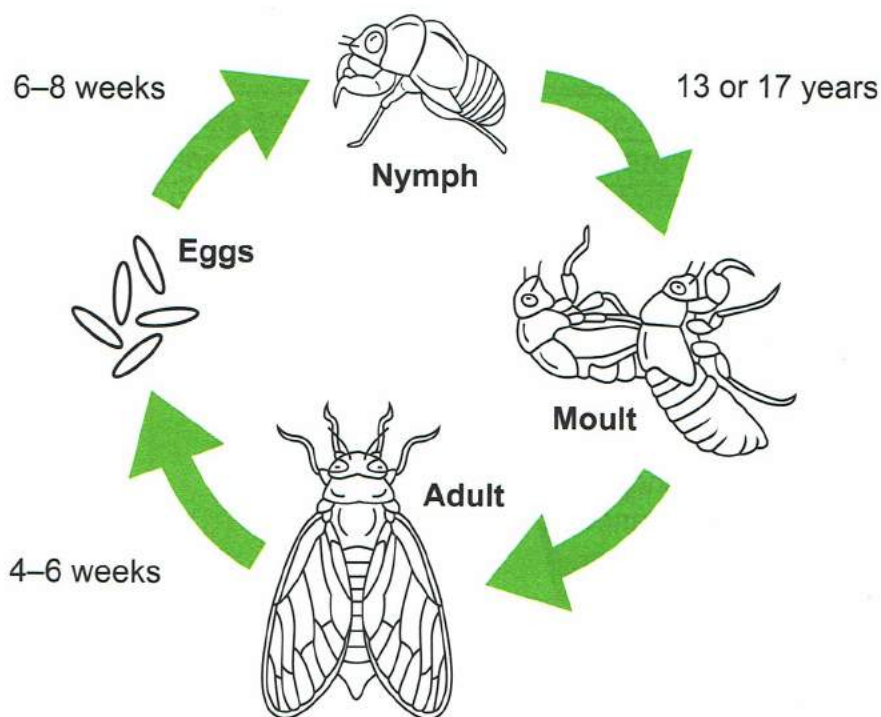
He then dropped them from the same height as shown.



Which sheet of paper will hit the ground first and for what reason?

	Paper that will hit the ground first	Reason
(A)	crushed sheet	It is heavier.
(B)	flat sheet	There is more air pushing it back.
(C)	crushed sheet	There is less air pushing it back.
(D)	both hit the ground at the same time	Both have the same mass.

12. Adult female cicadas in America lay eggs in the branches of trees. The eggs hatch into nymphs which then drop to the ground. The nymphs burrow underground and stay there for many years. Then they tunnel to the surface, crawl up the trees and shed their skin (moult). The adult female that comes out will mate and lay eggs.



How old is a cicada when it sheds its skin on a tree?

- (A) just born
(B) 4-6 weeks
(C) 6-8 weeks
(D) 13 or 17 years

13. Angela took these photos of four different snail shells, each one next to the same coin.



Max found a snail (T) in the yard and took a photo of it, also next to the same coin.



Which snail in Angela's photos is closest in size to Snail T?

- (A) Snail P
(B) Snail Q
(C) Snail R
(D) Snail S

14. Cali wanted to find out what substances look like when they are mixed with water.

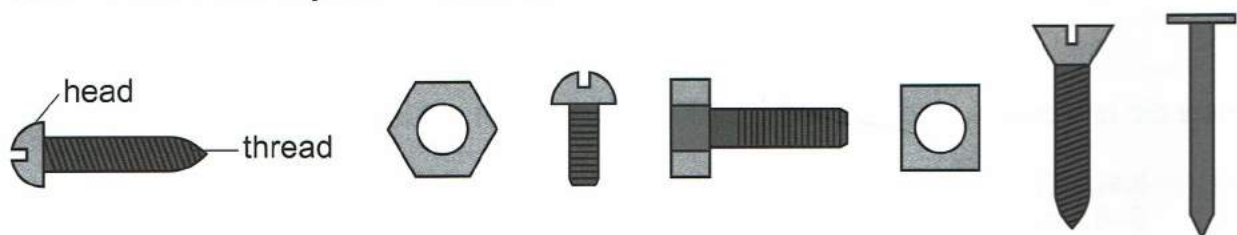
The table lists her results.

Substance mixed with water	What colour is the mixture?	Can you see through the mixture?
milk	white	no
mud	brown	no
salt	no colour	yes
sugar	no colour	yes
tea	brown	yes

Which of these did Cali have to keep the same to be able to compare her results?

- (A) the spoon she used to stir the mixture
- (B) the size of the glass to hold the mixture
- (C) the amount of substance added to the water
- (D) the colour of the substance added to the water

15. Jason found these objects in his toolbox.



He used this key to find the correct name for each object.

Question 1	Does it have a hole?	Yes	Go to Question 2.
		No	Go to Question 3.
Question 2	Does it have six sides?	Yes	It is a hexagonal nut.
		No	It is a square nut.
Question 3	Does it have a thread?	Yes	Go to Question 4.
		No	It is a nail.
Question 4	Does it have a rounded head?	Yes	It is a round-head screw.
		No	Go to Question 5.
Question 5	Does it have a pointy tip?	Yes	It is a countersunk screw.
		No	It is a hexagonal bolt.

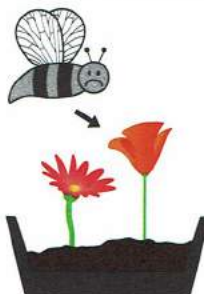
What is the name of this object?



- (A) hexagonal nut
- (B) nail
- (C) countersunk screw
- (D) hexagonal bolt

16. Scientists want to test whether parasites can spread from infected bees to healthy bees when they collect pollen from the same flowers.

The drawings show four steps of the test. They are not in the correct order.

W	X	Y	Z
Healthy bees collect pollen from the flowers.	Bees are tested to see if they carry the parasites.	Bees carrying parasites are removed from the flowers.	Bees carrying parasites collect pollen from healthy flowers.
			

In which order do the steps have to be carried out to make this a correct test?

- (A) X, W, Z, Y (B) Z, Y, W, X (C) W, Y, X, Z (D) Y, X, Z, W

17. Class 4A were learning about disposing of rubbish. The students put their waste either in the compost bin, the recycling bin or the general waste bin.

They recorded the amount of waste in each bin at the end of every week for 4 weeks.

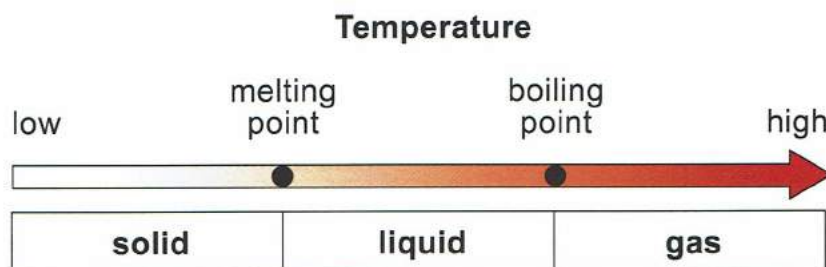
End of Week	Compost (grams)	Recycle bin (items)	General waste (items)
1	40	10	5
2	90	12	6
3	150	15	9
4	205	19	10

During which week did the amount in the compost bin increase the most?

- (A) Week 1 (B) Week 2
(C) Week 3 (D) Week 4

For questions 18 and 19 use the information below.

Things around us are made up of matter. Matter can exist in three main states: solid, liquid or gas. Matter may change state as its temperature changes. The temperature at which a solid becomes a liquid is called its melting point. The temperature at which a liquid becomes a gas is called its boiling point.



18. Which change results in a state of matter that can hold its own shape?

- | | |
|---------------------|---------------------|
| (A) liquid to solid | (B) liquid to gas |
| (C) gas to liquid | (D) solid to liquid |

19. Substance X has a melting point of 30°C and a boiling point of 120°C .

In what state is substance X at 25°C and in what state is it at 110°C ?

	at 25°C	at 110°C
(A)	solid	liquid
(B)	liquid	gas
(C)	liquid	solid
(D)	solid	gas

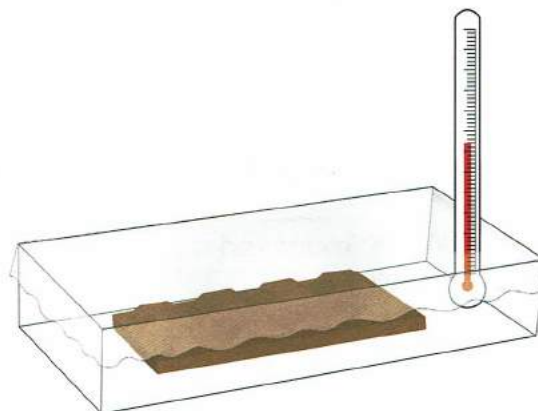
20. Andrew placed a chocolate bar in a clear box outside in the sunlight. He placed a thermometer in the box to measure the temperature inside the box.

He wrote down four statements before and during his investigation.

1. The chocolate will melt at 35°C .
2. The chocolate lost its crystal structure.
3. The chocolate began to melt at 31°C .
4. The box trapped heat inside it.

Which statement was an observation?

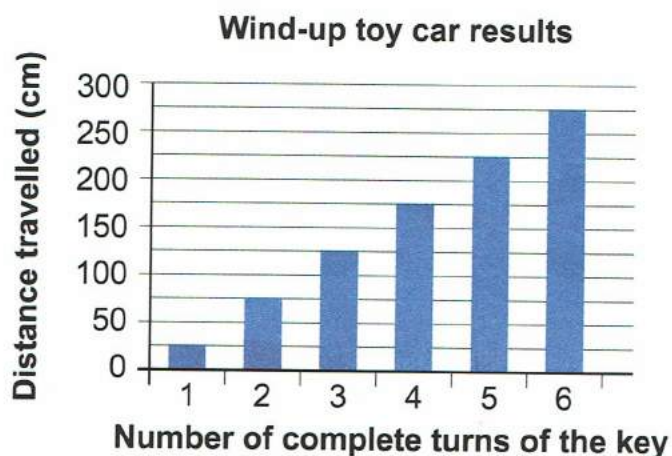
- | | |
|-------|-------|
| (A) 1 | (B) 2 |
| (C) 3 | (D) 4 |



21. Wind-up toy cars move forwards when you wind them up with a key and let them go.

Frank carried out an investigation to find out if the number of times he turns the key affects the distance the car travels.

The graph shows Frank's results.



Which two things did Frank have to keep the same to make his investigation fair?

- (A) the car and the distance it travelled
- (B) the car and the surface on which it moved
- (C) the distance the car travelled and the surface on which it moved
- (D) the number of turns of the key and the surface on which the car moved

22. Many things can affect how well plants grow. These include the nitrogen content of the soil and its acidity, given by its pH. Two brands of fertiliser advertise the following on the packages.



- ✓ Raises pH
- ✓ Nitrogen Rich!



- ✓ Lowers pH
- ✓ Nitrogen Rich!

Farmer Wendy tested the soil of her vegetable garden. As a result she decided to use *Grow & Grow* to improve her soil.

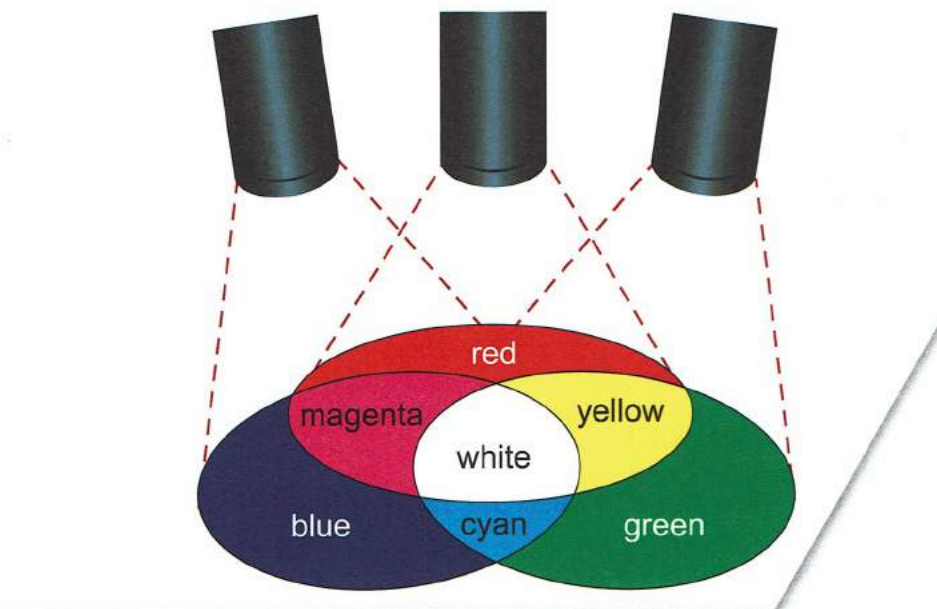
What did Wendy find from her soil test that made her decide to use *Grow & Grow*?

The soil had

- (A) low nitrogen and high pH.
- (B) high nitrogen and low pH.
- (C) low nitrogen and low pH.
- (D) high nitrogen and high pH.

23. There are three colours of light called primary colours. When you mix two primary colours of light, you produce light with a secondary colour. Mixing all three primary colours in the right proportion produces white light.

Kate had three torches that each produced a different primary colour of light. She shone the torches onto a flat white surface, so their light beams overlapped.



What colour of light do you get when you mix blue light and yellow light?

- (A) red (B) white (C) magenta (D) cyan

24. Matt was walking along the sand when he saw these footprints.



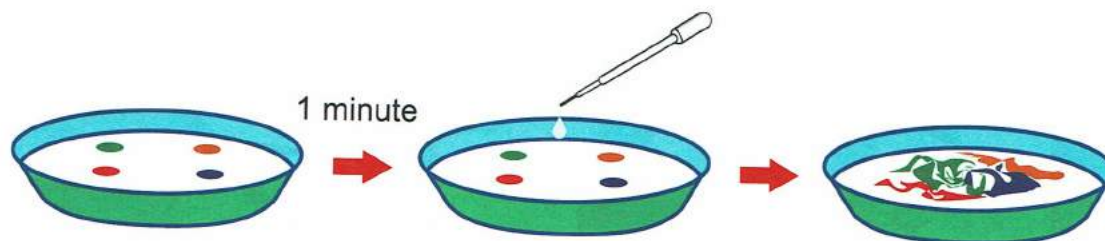
Which option shows the correct order of footprints on the sand, from first footprint to last footprint?

- (A) cat, horse, human, bird (B) bird, human, horse, cat
(C) horse, human, bird, cat (D) human, bird, horse, cat

25. Mrs Black showed this science experiment to her Year 3 class.

She put a small drop of four food colours onto the surface of some milk in a dish.

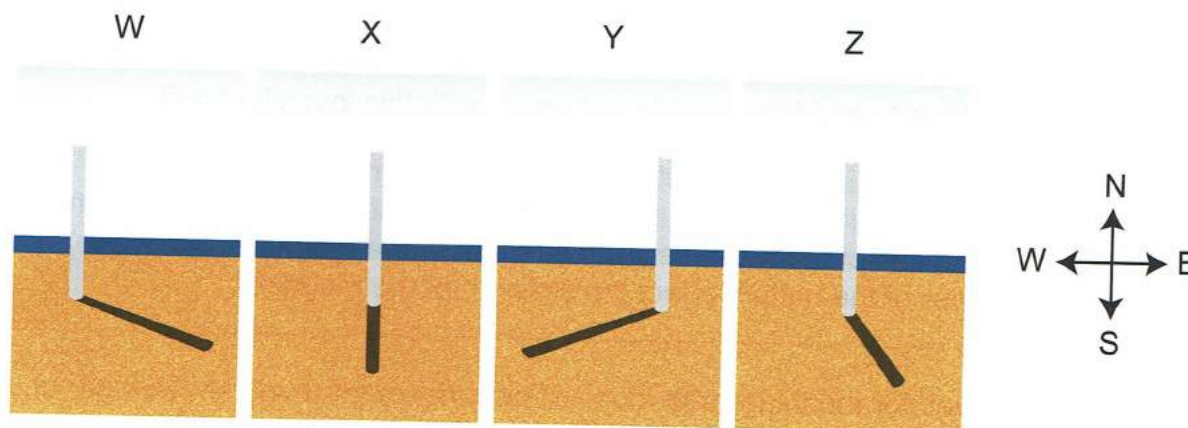
After 1 minute she carefully added a drop of detergent onto the middle of the dish. The colours immediately started moving and mixing with each other forming colourful patterns on the surface of the milk.



Which statement is supported by these observations?

- (A) The milk surface is constantly moving, making the colours mix.
- (B) The milk surface is smooth, making the colours slide around.
- (C) The colours mix with the detergent, making the milk surface move and mix.
- (D) The detergent mixes with milk, making the milk surface move and the colours mix.

26. These drawings show the shadows cast by a stick in the sand at different times during the day.



The Sun rises in the East and sets in the West. Drawing W shows the shadow that the stick made at 6:00 pm.

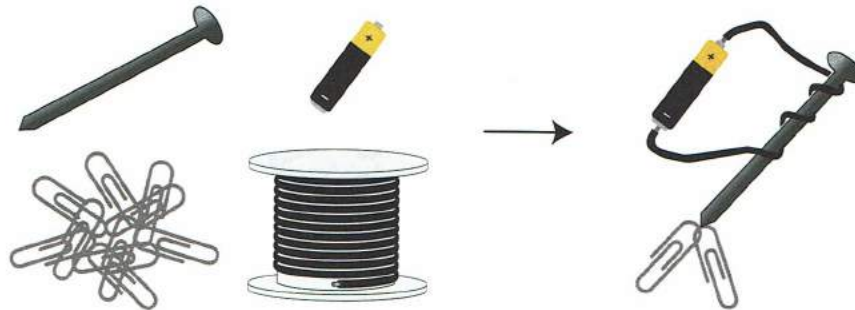
Which of these matches the shadows X, Y and Z with the time they were drawn?

	X	Y	Z
(A)	6 am	10 am	3 pm
(B)	12 pm (noon)	10 am	6 am
(C)	12 pm (noon)	6 am	3 pm
(D)	12 am	6 am	10 am

For questions 27 and 28 use the information below.

Xiao Ming read in a book that electricity can be used to turn a piece of iron into a magnet. He decided to test this idea.

He carried out his investigation using a large iron nail, some plastic-coated wire, a brand new battery and some paperclips. These drawings show what he did.



27. Which option correctly identifies whether this was a fair test and gives a correct reason?

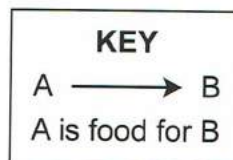
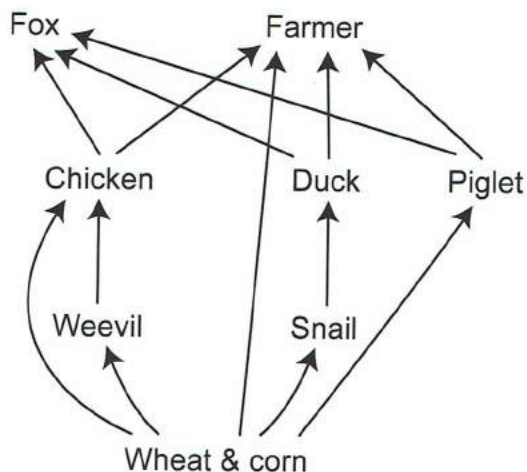
	Is it a fair test?	What is the reason?
(A)	yes	The paperclips stuck to the nail with electricity around it.
(B)	yes	Electricity made the nail sticky.
(C)	no	He did not test if the nail alone would pick up the paperclips.
(D)	no	The battery is not powerful enough.

28. Xiao Ming wanted to carry out another investigation with the materials he had.

Which of these questions would he be able to answer with this investigation?

- (A) Does the size of the iron nail affect the strength of the magnet?
- (B) Does the number of loops of wire around the nail affect the strength of the magnet?
- (C) Does the thickness of the wire affect the strength of the magnet?
- (D) Does the power of the battery affect the strength of the magnet?

29. Ruby drew a food web for the things she found at her uncle's farm and showed her uncle.



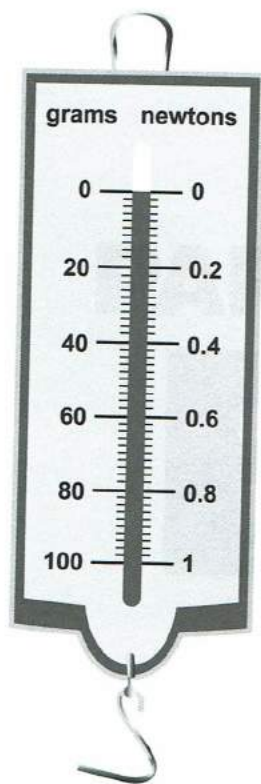
One day, Ruby's uncle said "Oh no, the fox ate it! But at least the chicken will have more food!"

Based on her food web, what had the fox most likely eaten?

- (A) piglet
- (B) snail
- (C) duck
- (D) weevil

30. A spring balance can be used to measure the mass and the weight of an object. It measures mass in grams and weight in newtons.

Ali used a spring balance to find the mass of some keys.



key 1



key 2



key 3



key 4

The table shows his results.

Key(s)	Mass (grams)
1	14
1+2	27
1+2+3	42
1+2+3+4	57

Which two keys have the same mass?

- (A) Key 1 and Key 2
(B) Key 2 and Key 3
(C) Key 2 and Key 4
(D) Key 3 and Key 4

Acknowledgment

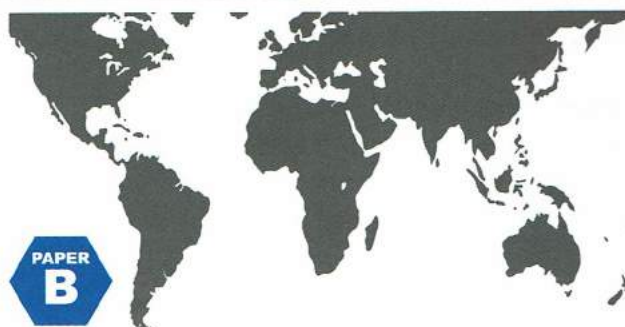
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Sources

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The following year levels should sit **THIS Paper**:

Australia	Year 4
Brunei	Primary 4
Indonesia	Year 5
Malaysia	Standard 4
New Zealand	Year 5
Pacific Region	Year 5
Singapore	Primary 3
South Africa	Grade 4



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Science Year 4 _ Paper B 2016

Question Number	Area Assessed	Question Description	Correct Answer
1	Observing/Measuring	Match a diagram with a written description	D
2	Interpreting	Classify skeletal parts using a diagram and given information	A
3	Predicting/Concluding	Infer the reason for the change of water level in a river	D
4	Observing/Measuring	Use information in a diagram to calculate the difference between two temperature readings	B
5	Interpreting	Interpret information in a table	A
6	Interpreting	Obtain information from a table	D
7	Reasoning/Problem solving	Predict the effect of a number of forces applied to an object at the same time	A
8	Interpreting	Interpret a diagram dealing with an astronomical phenomenon	B
9	Predicting/Concluding	Identify a reversible change using given information	B
10	Predicting/Concluding	Use a dichotomous key	C
11	Predicting/Concluding	Recognise the effect of air resistance on objects with a different surface area	C
12	Interpreting	Interpret a diagram related to the life cycle of an organism	D
13	Observing/Measuring	Compare the size of objects using a reference object	C
14	Investigating	Identify a feature of a fair test	C
15	Observing/Measuring	Determine the name of an object using a dichotomous key	D
16	Investigating	Order the steps of an investigation to make it a valid test	B
17	Interpreting	Interpret information in a table	C
18	Predicting/Concluding	Obtain information from a diagram	A
19	Reasoning/Problem solving	Use information provided to find the state of matter of a substance at different temperatures	A
20	Investigating	Identify a statement that is an observation	C
21	Investigating	Identify the controlled variable in an investigation	B
22	Reasoning/Problem solving	Deduce the properties of a substance	A
23	Reasoning/Problem solving	Determine the result of mixing two colours of light	B
24	Reasoning/Problem solving	Locate a fossil using map coordinates	B
25	Investigating	Draw a conclusion based on the results of an investigation	D
26	Reasoning/Problem solving	Deduce the position of shadows during a day	C
27	Investigating	Identify the control in an investigation	C
28	Investigating	Identify the question that can be answered with a given set of materials	B
29	Predicting/Concluding	Use a drawing of a food web to draw a conclusion	A
30	Reasoning/Problem solving	Determine which objects have the same mass using data in a table	D

